

Exercices on session *Statistics with R session*

2/3

Load the packages of the *tidyverse* and *ggeffects*

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(ggeffects)
```

Data on vascular plants in the British Islands

The Arrhenius relationship ([1921](#)) predicts that the number of species in an ecosystem increases with its area to the power z according to the equation

$$S = cA^z$$

where $S(A)$ is the number of species observed on area A , c is a constant that depends on the ecosystems and the taxon considered, and z is the parameter of interest.

We are going to explore this relationship using data on the number of vascular species in the British Islands (Johnson et Simberloff, 1974), available [here](#).

- Load the data as a tibble and look at their structure

```
dt <- read_csv("data/britain_species.csv")
```

```
Rows: 42 Columns: 7
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): island
```

```
dbl (6): area, elevation, soil_types, latitude, dist_britain, species
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
summary(dt)
```

	island	area	elevation	soil_types
Length	:42	Min. : 0.50	Min. : 15.0	Min. : 1.000
N.unique	:42	1st Qu.: 14.00	1st Qu.: 129.5	1st Qu.: 1.000
N.blank	: 0	Median : 58.15	Median : 232.0	Median : 2.000
Min.nchar	: 3	Mean : 5740.12	Mean : 355.0	Mean : 2.643
Max.nchar	:11	3rd Qu.: 380.38	3rd Qu.: 470.2	3rd Qu.: 3.000
		Max. :229849.80	Max. :1343.0	Max. :16.000
	latitude	dist_britain	species	
Min.	:50.70	Min. : 0.000	Min. : 62.0	
1st Qu.	:56.12	1st Qu.: 9.175	1st Qu.: 174.8	
Median	:57.05	Median : 30.050	Median : 383.0	
Mean	:57.28	Mean : 63.157	Mean : 392.2	
3rd Qu.	:59.25	3rd Qu.: 81.075	3rd Qu.: 465.8	
Max.	:60.80	Max. :258.100	Max. :1666.0	

The data contains the number of species of vascular plants (variable *species*) for different *island*. The other variables give some characteristics of these islands, including the *area* in km².

```
dt <- dt %>% filter(island != "Britain")
```

Simple linear regression

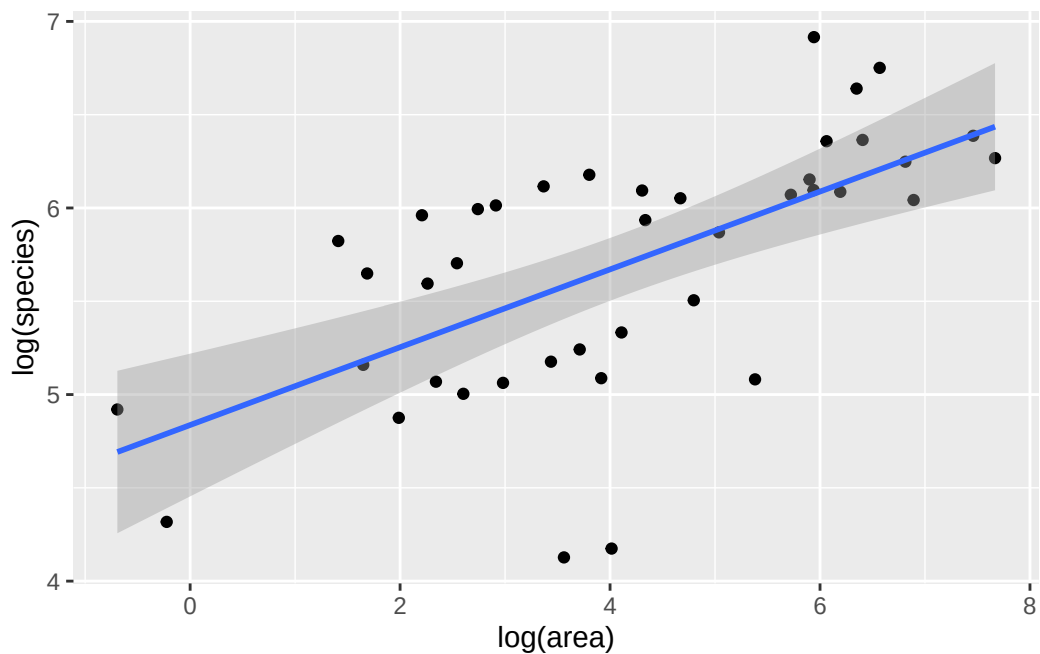
The Arrhenius relationship can be linearised by taking the logarithm of S and A :

$$\log(S) = \beta_0 + \beta_1 \log(A)$$

- Explore graphically the relationship between the log-transformed area and the log-transformed number of species

```
dt %>%  
  ggplot(aes(x = log(area), y = log(species))) +  
  geom_point() +  
  geom_smooth(method = "lm")
```

`geom\_smooth()` using formula = 'y ~ x'



- Fit the following log-log model, look at the model results and interpret them

$$\log(S) \sim \mathcal{N}(\beta_0 + \beta_1 \times \log(A), \sigma^2)$$

```
mod_1 <- lm(log(species) ~ log(area), data = dt)
summary(mod_1)
```

Call:

```
lm(formula = log(species) ~ log(area), data = dt)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.49944	-0.36085	0.02073	0.35908	0.83992

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.83657	0.18915	25.570	< 2e-16 ***
log(area)	0.20855	0.04129	5.052	1.06e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5318 on 39 degrees of freedom

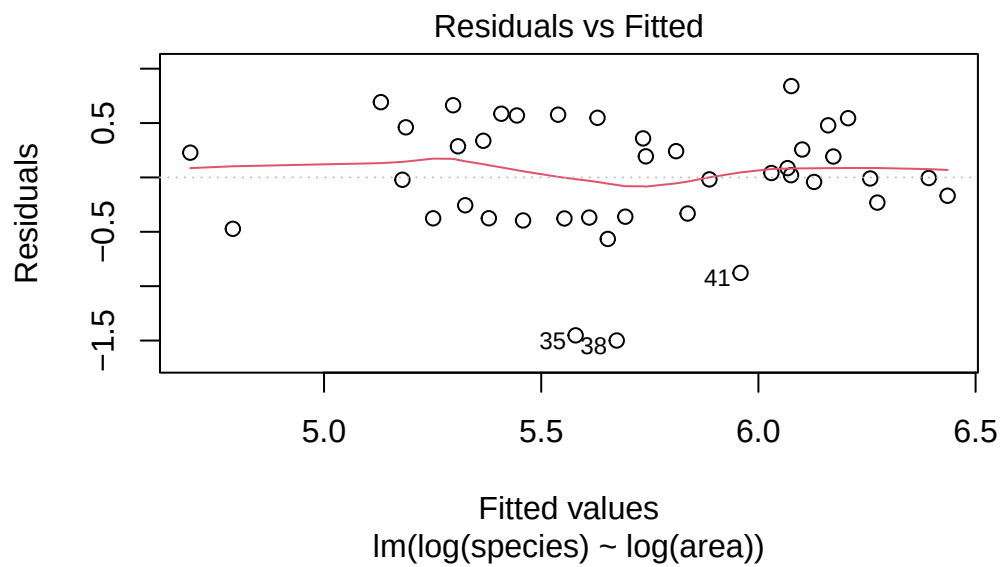
Multiple R-squared: 0.3955, Adjusted R-squared: 0.38

F-statistic: 25.52 on 1 and 39 DF, p-value: 1.064e-05

```
# interpretation
# significant positive effect of the log(area)
# moderate goodness of fit (R2)
```

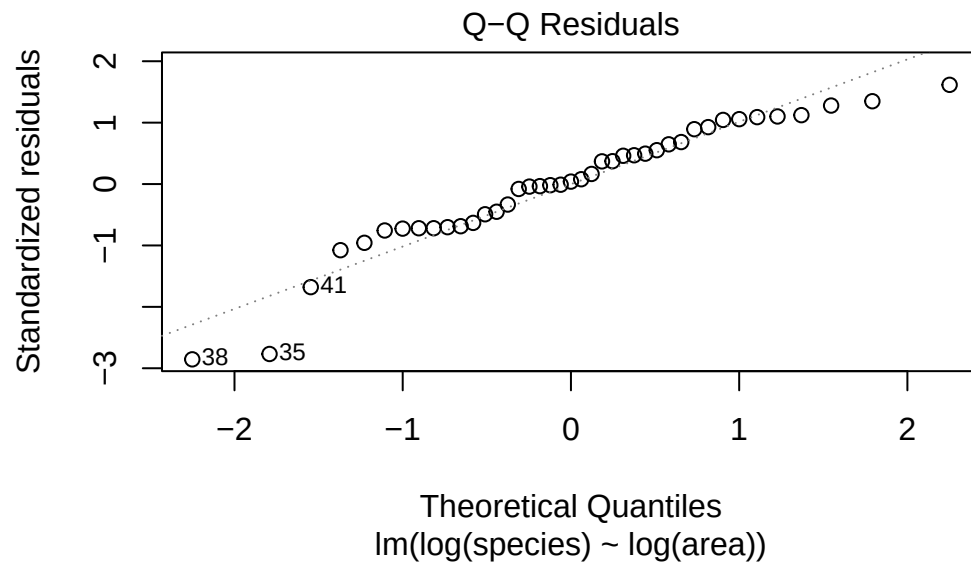
- Do the model validation

```
plot(mod_1, which = 1)
```



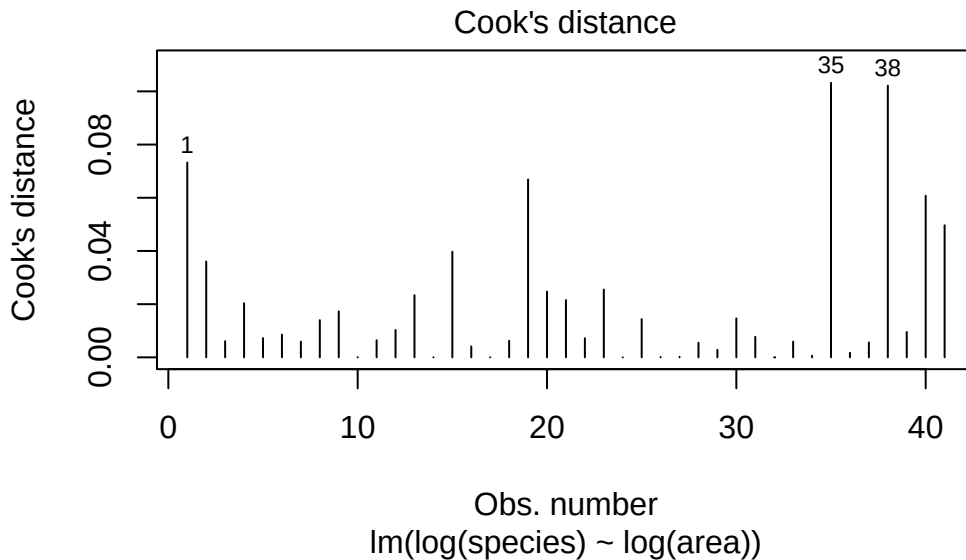
```
# interpretation
# homoscedasticity: not too bad
# residuals independant of fitted values
```

```
plot(mod_1, which = 2)
```



```
# interpretation  
# normality of the residuals... not perfect
```

```
plot(mod_1, which = 4)
```



```
# interpretation
# No very influential observations, the distance of Cook are small
```

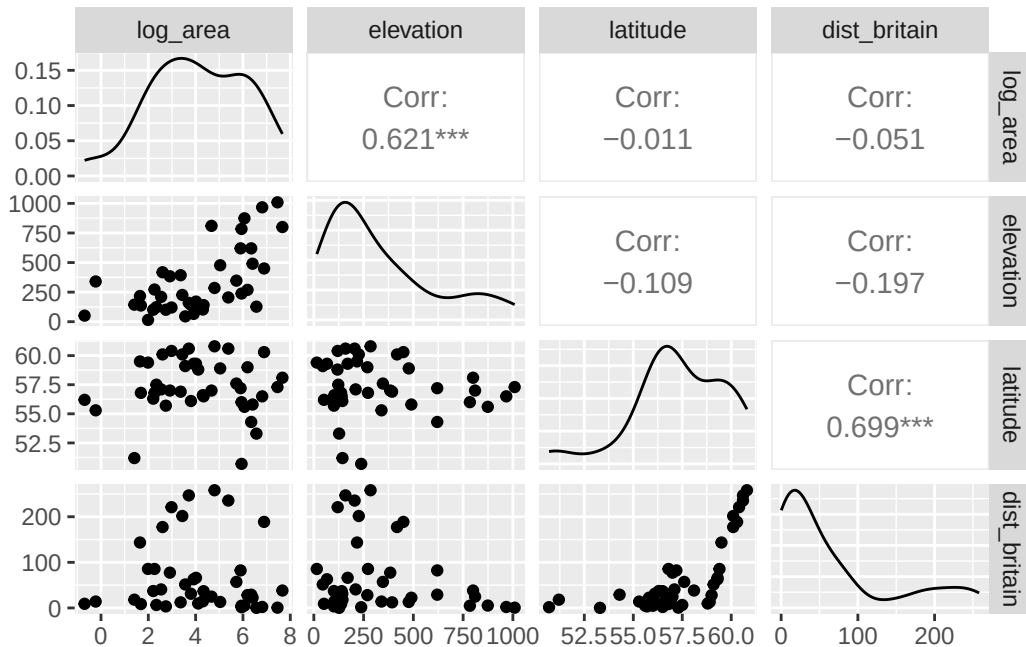
Multiple linear regression

The goodness of fit of the previous model could probably be improved. As the variables *elevation*, *latitude*, *dist_britain* and *soil_types* could also be influencing the number of species, we could add them as explanatory variables in our model.

- Let's first explore graphically the correlation between all pairs of numerical variables, *area* (log-transformed), *elevation*, *latitude*, *dist_britain*, using the package GGally. What do you conclude?

```
dt <- dt %>%
  mutate(log_area = log(area))

GGally::ggpairs(dt,
  columns = c("log_area", "elevation",
              "latitude", "dist_britain"),
  progress = FALSE) # no progress bar
```

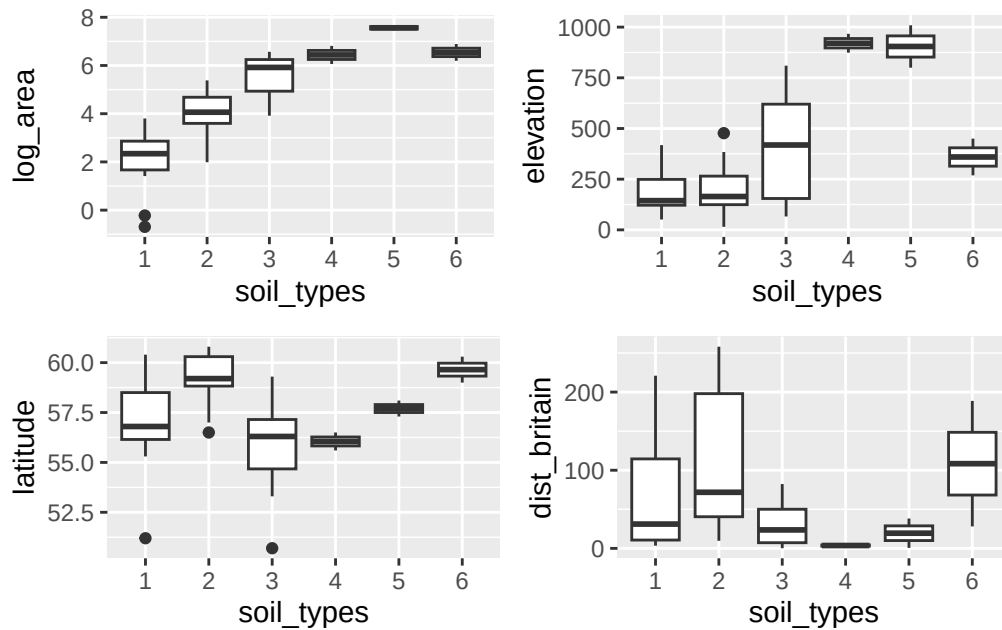


```
# interpretation
# log(area) and elevation correlated
# dist_britain and latitude correlated
# => don't include them all (and too many for the data we have)
```

- Let's also make boxplots to explore graphically the relationship between these 4 variables and the type of soil (don't forget to transform the variable *soil_types* to a factor). What do you conclude?

```
dt <- dt %>% mutate(soil_types = as.factor(soil_types))

g1 <- dt %>%
  ggplot(aes(x= soil_types, y= log_area)) + geom_boxplot()
g2 <- dt %>%
  ggplot(aes(x= soil_types, y= elevation)) + geom_boxplot()
g3 <- dt %>%
  ggplot(aes(x= soil_types, y= latitude)) + geom_boxplot()
g4 <- dt %>%
  ggplot(aes(x= soil_types, y= dist_britain)) + geom_boxplot()
library(patchwork)
(g1 + g2) / (g3 + g4)
```

```
# interpretation
# soil type too correlated with log_area...
# and some soil types poorly represented
```

- Fit the following model (without standardising the variables), look at the model results and interpret them

$$\log(S) \sim \mathcal{N}(\beta_0 + \beta_1 \times \log(A) + \beta_2 \times \text{latitude}, \sigma^2)$$

```
mod_2 <- lm(log(species) ~ log(area) + latitude, data = dt)
summary(mod_2)
```

Call:

```
lm(formula = log(species) ~ log(area) + latitude, data = dt)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.19645	-0.13566	0.04961	0.22469	0.51189

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.79032	1.48513	9.286	2.57e-11 ***
log(area)	0.20662	0.02984	6.924	3.12e-08 ***
latitude	-0.15597	0.02576	-6.055	4.79e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3843 on 38 degrees of freedom

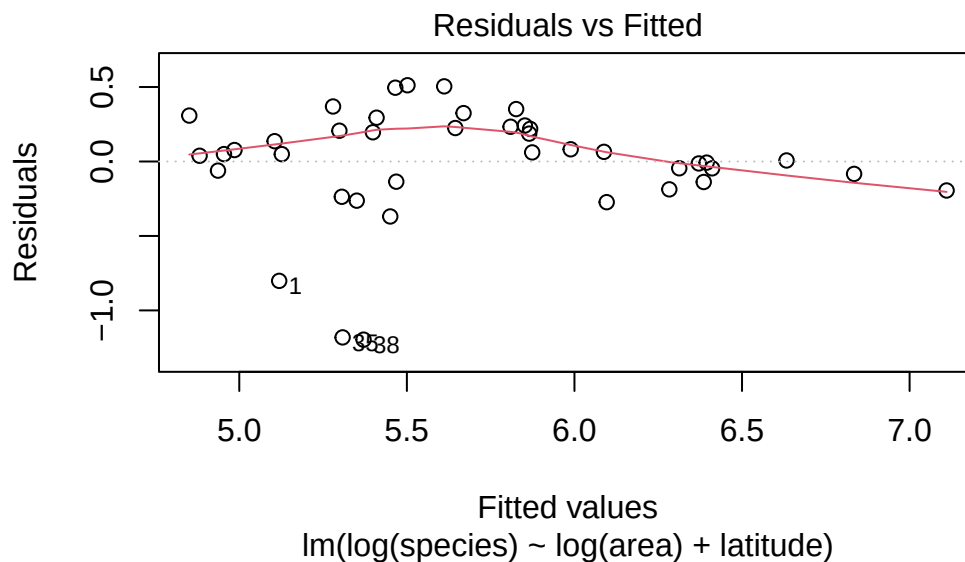
Multiple R-squared: 0.6923, Adjusted R-squared: 0.6761

F-statistic: 42.75 on 2 and 38 DF, p-value: 1.878e-10

```
# interpretation
# Goodness of fit (R2) much better
# log(area) and latitude both have a significant effect
# number of species increases with area
# number of species decreases with latitude
```

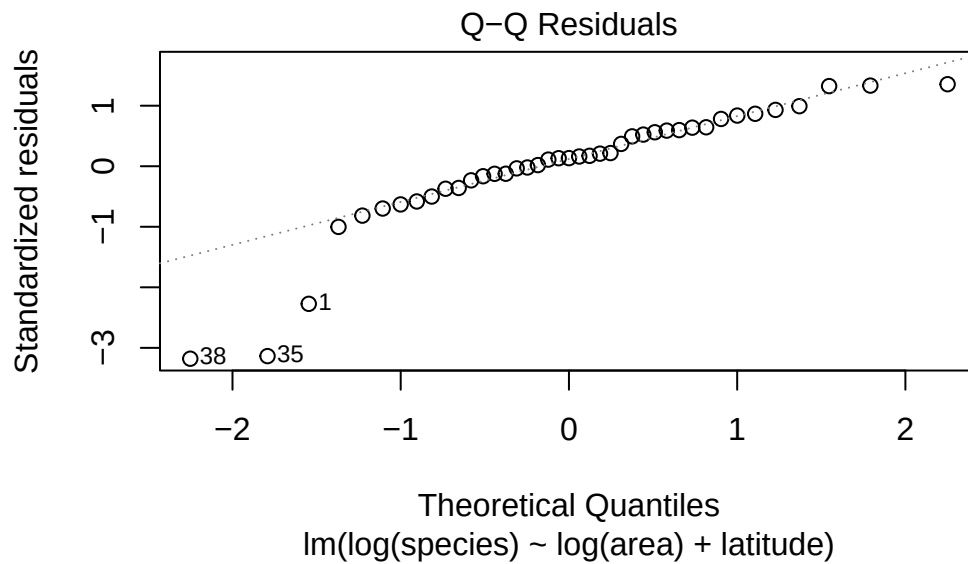
- Do the model validation

```
plot(mod_2, which = 1)
```



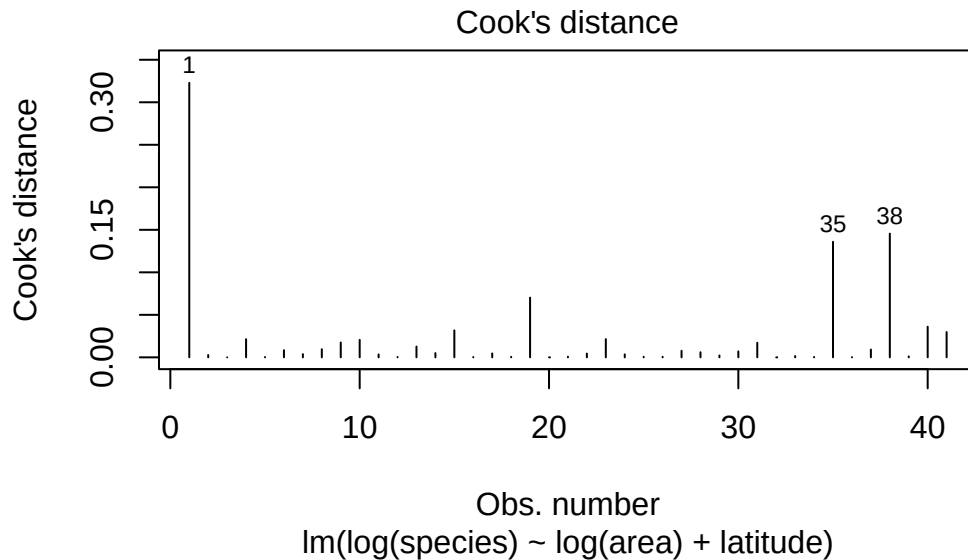
```
# interpretation
# homoscedasticity: not too bad
# residuals independent of fitted values
# three obs concerning
```

```
plot(mod_2, which = 2)
```



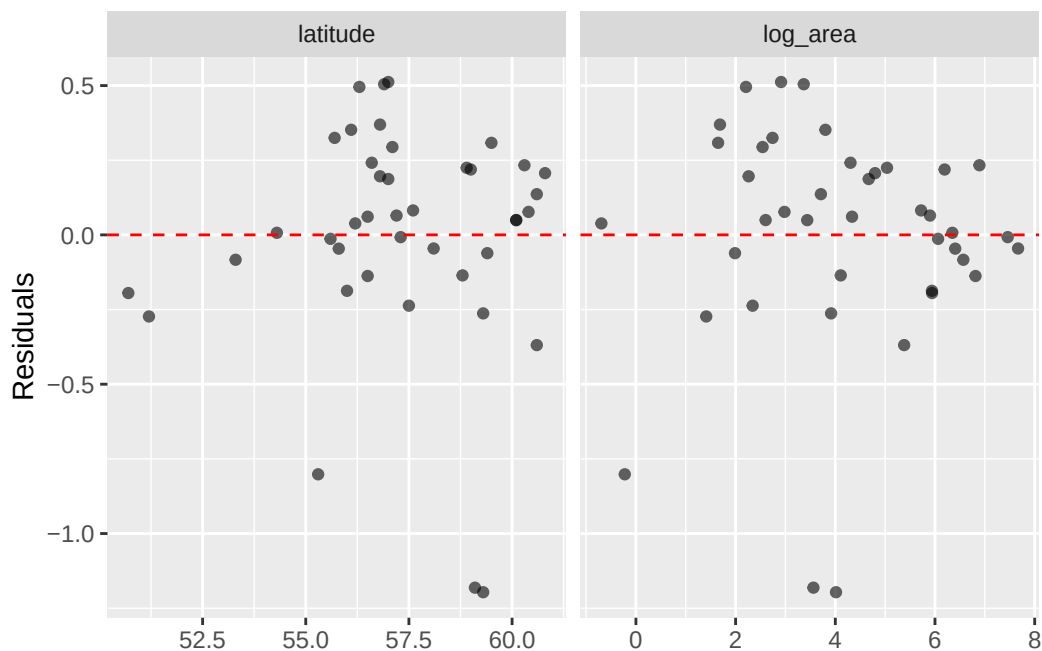
```
# interpretation
# normality of the residuals
# three obs concerning
```

```
plot(mod_2, which = 4)
```



```
# interpretation
# No very influential observations, but some obs concerning
```

```
dt %>%
  mutate(resid = residuals(mod_2)) %>% # add the residuals
  pivot_longer(cols = c(log_area, latitude),
               names_to = "predictor",
               values_to = "value") %>% # change to long table
  ggplot(aes(x = value, y = resid)) +
  geom_point(alpha = 0.6) +
  geom_hline(yintercept = 0, linetype = "dashed", col = "red") +
  facet_wrap(~ predictor, scales = "free_x", ncol = 2) +
  labs(x = NULL, y = "Residuals")
```



```
# interpretation
# ok
```

- Check the VIF (from library *car*) to check for the absence of collinearity between the explanatory variables

```
car::vif(mod_2)
```

```
log(area)  latitude
1.000114   1.000114
```

```
# interpretation
# ok
```

Presenting the result of the best model

- Compare the AIC of the two models. Which is the best one?

```
list_mod <- list(mod_1, mod_2)
AICcmodavg::aictab(list_mod)
```

Characteristic	Beta	95% CI	p-value
(Intercept)	14	11, 17	<0.001
log(area)	0.21	0.15, 0.27	<0.001
latitude	-0.16	-0.21, -0.10	<0.001

Abbreviation: CI = Confidence Interval

Warning in aictab.AIClm(list_mod):

Model names have been supplied automatically in the table

Model selection based on AICc:

	K	AICc	Delta_AICc	AICcWt	Cum.Wt	LL
Mod2	4	43.94	0.00	1	1	-17.41
Mod1	3	69.16	25.23	0	1	-31.26

```
# best model is model 2
```

- Present the coefficients in a table

```
gtsummary::tbl_regression(mod_2, intercept = TRUE)
```

```
# in the corrected exercise, the table might be above
```

- Present the result graphically: first, represent the predicted relationship between A and S, setting the latitude to its mean

```
preds4area <- predict_response(mod_2, terms = "area")
```

Model has log-transformed response. Back-transforming predictions to original response scale. Standard errors are still on the transformed scale.

```
preds4area
```

```
# Predicted values of species
```

area	Predicted	95% CI
0	0.00	
260	400.81	345.12, 465.49
540	466.15	389.76, 557.51
800	505.59	415.15, 615.72
1060	535.86	434.04, 661.55
1340	562.45	450.28, 702.56
1600	583.44	462.87, 735.40
2140	619.57	484.16, 792.85

```
Adjusted for:
```

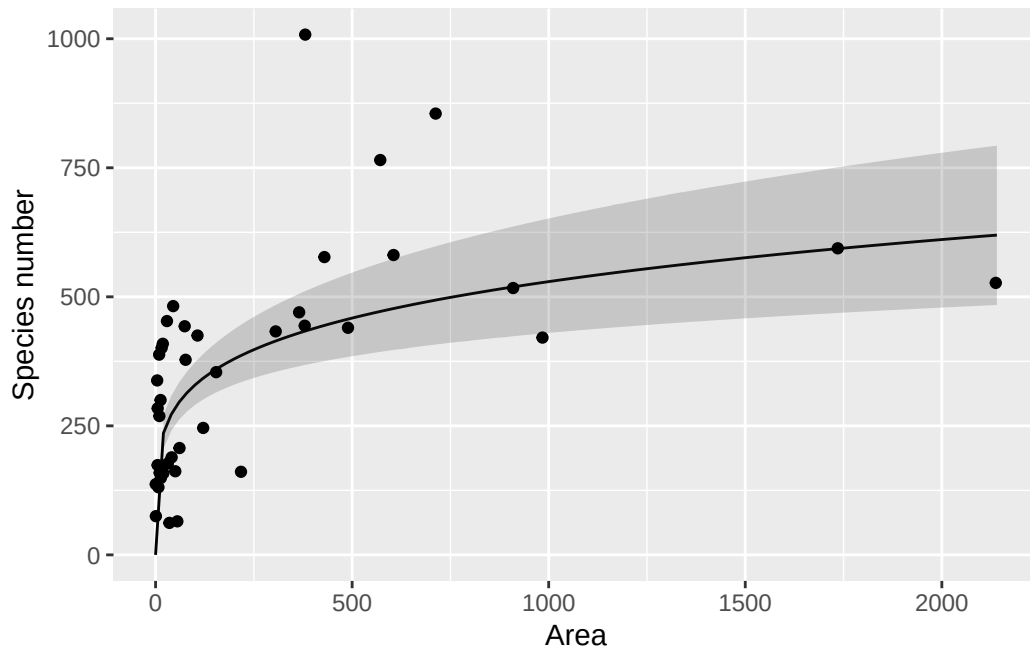
```
* latitude = 57.36
```

Not all rows are shown in the output. Use ``print(..., n = Inf)`` to show all rows.

```
# str(preds4area)
```

```
ggplot(preds4area, aes(x = x, y = predicted)) +  
  geom_line() +  
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.2) +  
  geom_point(data = dt, aes(x = area, y = species)) +  
  labs(x = "Area", y = "Species number")
```

Warning: Removed 1 row containing missing values or values outside the scale range (``geom_ribbon()``).



- then the predicted relationship between latitude and S, setting the area to its mean

```
preds4lat <- predict_response(mod_2, terms = "latitude")
```

Model has log-transformed response. Back-transforming predictions to original response scale. Standard errors are still on the transformed scale.

```
preds4lat
```

Predicted values of species

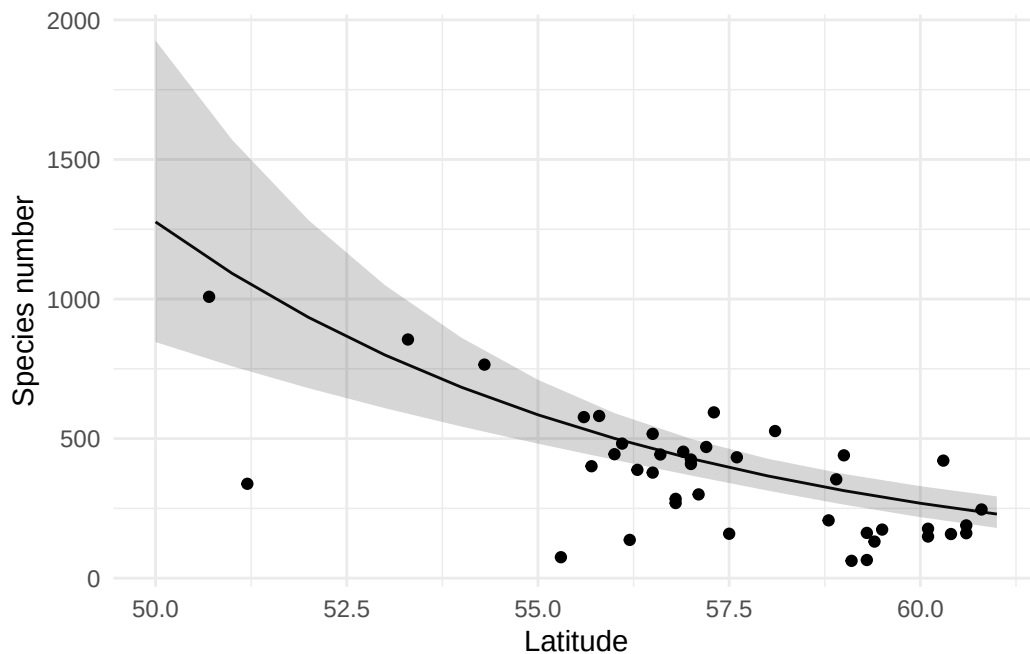
latitude	Predicted	95% CI
50	1276.24	845.68, 1926.01
51	1091.93	759.11, 1570.66
53	799.32	608.83, 1049.42
54	683.89	542.98, 861.37
55	585.12	481.74, 710.70
57	428.33	367.75, 498.88
58	366.47	313.74, 428.06
61	229.52	179.87, 292.88

Adjusted for:
* area = 274.03

Not all rows are shown in the output. Use ``print(..., n = Inf)`` to show all rows.

```
# str(preds4lat)
```

```
ggplot(preds4lat, aes(x = x, y = predicted)) +  
  geom_line() +  
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.2) +  
  geom_point(data = dt, aes(x = latitude, y = species)) +  
  labs(x = "Latitude", y = "Species number") +  
  theme_minimal()
```



Acknowledgments

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